

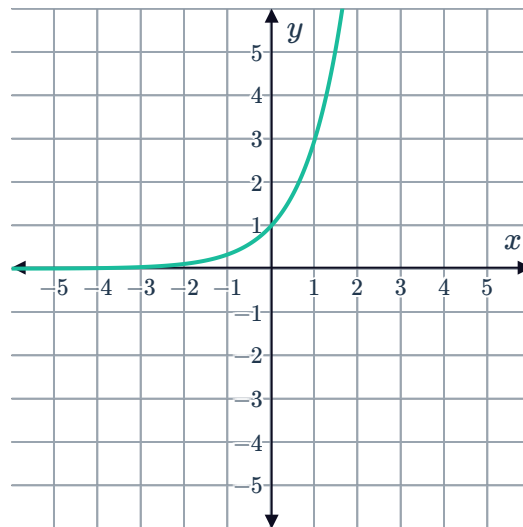
Transformations of exponential functions



1 The graph of $y = 2^x$ is translated down by 7 units, state its new equation.

2 Consider the given graph of $y = 3^x$.

- Describe a transformation of the graph of $y = 3^x$ that would obtain $y = 3^x - 4$.
- Sketch the graph of $y = 3^x - 4$.



3 Consider the function $y = 9^x + 5$.

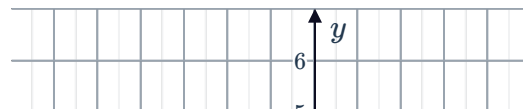
- What value is 9^x always greater than?
- Now, what value is $9^x + 5$ always greater than?
- How many x -intercepts does $y = 9^x + 5$ have?
- State the equation of the asymptote of the curve $y = 9^x + 5$.

4 Consider the functions $y = 2^x$ and $y = 2^x - 2$.

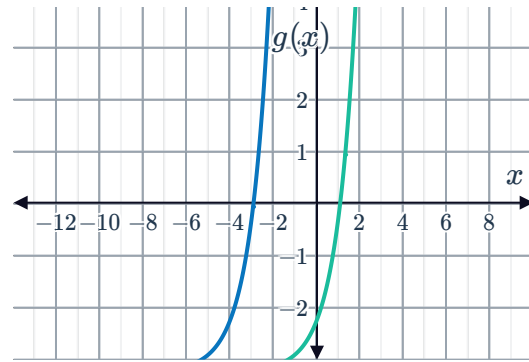
- Find the y -intercept of $y = 2^x$.
- Now, determine the y -intercept of $y = 2^x - 2$.
- State the horizontal asymptote of $y = 2^x$.
- Now, determine the horizontal asymptote of $y = 2^x - 2$.

5 Sarah graphs the function $f(x) = 2^x + 6$. She then translated the graph down 9 units to create the function $g(x)$. State the equation of $g(x)$.

6 Consider the given graphs of $f(x) = 3^x$ and $g(x)$:

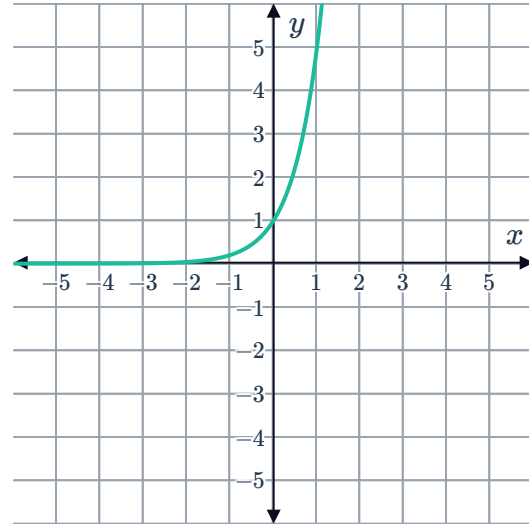


- a Describe a transformation that can be used to obtain $g(x)$ from $f(x)$.
- b State the equation of $g(x)$.



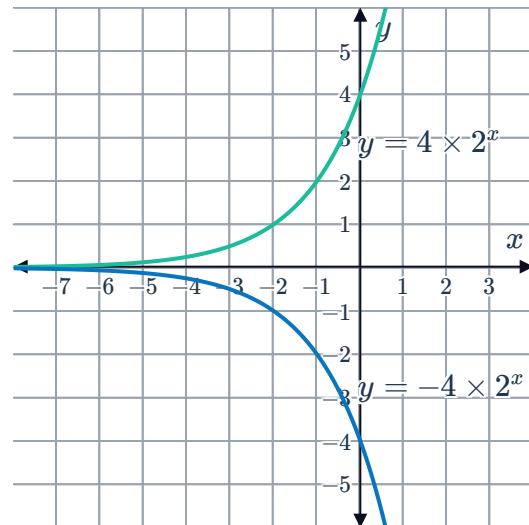
7 Consider the given graph of $y = 5^x$:

- a Describe a transformation of the graph of $y = 5^x$ that would obtain $y = -5^x$.
- b Sketch the graph of $y = -5^x$.



8 Consider the given graphs of $y = 4 \times 2^x$ and $y = -4 \times 2^x$:

- a Find the value of y in $y = 4 \times 2^x$ when $x = 0$.
- b Describe a transformation of the graph of $y = 4 \times 2^x$ that would obtain $y = -4 \times 2^x$.
- c The number of bacteria over time is to be modeled by an exponential function, with x representing time and y representing the number of bacteria. If the bacteria are increasing, which of the two models should be used?



9 Of the two functions $y = 2^x$ and $y = 3 \times 2^x$, which is increasing more rapidly for $x > 0$?

- a Find the y -intercept of the curve.
- b Is the function value ever negative?
- c As x approaches infinity, what value does y approach?
- d Sketch the graph of $y = 4(2^x)$.



11 Consider the original graph $y = 3^x$. The function values of the graph are multiplied by 2 to form a new graph.

- a For each point on the original graph, find the point on the new graph:

Point on original graph	$\left(-1, \frac{1}{3}\right)$	$(0, 1)$	$(1, 3)$	$(2, 9)$
Point on new graph	$\left(-1, \frac{2}{3}\right)$	$(0, 2)$	$(1, 6)$	$(2, 18)$

- b State the equation of the new graph.
- c Sketch the graph of $y = 3^x$ and $y = 2 \times 3^x$ on the same coordinate plane.
- d For negative x -values, is the graph of $y = 2 \times 3^x$ above or below $y = 3^x$?
- e For positive x -values, is the graph of $y = 2 \times 3^x$ above or below 3^x ?

12 Consider the function $y = -2.5 \times 4^x$.

- a Is $y = -2.5 \times 4^x$ an increasing or decreasing function?
- b As x approaches $-\infty$, what value does y approach?
- c As x approaches ∞ , what value does y approach?
- d State the y -intercept of the graph.